Likhith Nattuva

Sterling, VA 20164 | +1 (848) 336-9598 | likhith2201@gmail.com | GitHub

Software Engineer — Data Engineer — Data Analyst — Python — Java — SQL — AWS — AI/ML

Professional Summary

Computer Science graduate student with hands-on academic, internship, and portfolio project experience across software engineering, data analysis, data engineering, cloud data workflows, and AI/ML concepts. Skilled in Python, SQL, Java, JavaScript/TypeScript, React, Node.js, REST APIs, relational databases, AWS S3/Redshift concepts, ETL/ELT pipelines, data cleaning, data validation, testing, debugging, and technical documentation. Strong foundation in data structures, algorithms, database design, SDLC, reporting, and problem-solving, with experience translating requirements into reliable applications, analytics-ready datasets, and documented technical solutions.

Education

Harrisburg University — Harrisburg, PA
MS in Computer Information Science, **GPA: 3.9/4** — Mar 2025 – Jun 2026

K L University — Hyderabad, India
BTech in Computer Science and Engineering, **CGPA: 8.24/10** — Jun 2019 – May 2023

Technical Skills

Programming: Python, SQL, Java, JavaScript, TypeScript, C++, C, Scala
Software & Web: React, Node.js, Express.js, REST APIs, HTML5, CSS, Bootstrap, Django, Prisma ORM, responsive UI concepts
Data Engineering & Cloud: AWS S3, AWS Redshift, ETL/ELT, data pipelines, data lake concepts, data warehouse concepts, PySpark, Hive, Parquet, CSV, JSON
Databases & Data Quality: PostgreSQL, MySQL, OracleDB, MongoDB, schema design, data modeling, joins, aggregations, views, indexes, source-to-target checks, deduplication, null handling
Analytics & AI/ML: Pandas, NumPy, Matplotlib, basic statistics, trend analysis, reporting, dashboards, TensorFlow, Keras, OpenCV, CNN, model evaluation, AI-assisted development familiarity
Tools & Practices: Git, Jupyter Notebook, Docker concepts, Cucumber.js, Playwright, unit/acceptance testing, debugging, SDLC, Agile concepts, technical documentation

Professional Experience

Web Design Intern
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- Developed responsive web pages using HTML and CSS, translating requirements into structured layouts and functional page designs.
- Built foundational experience in SDLC, debugging, application structure, browser testing, and technical documentation.
- Collaborated in a learning-focused technical environment to understand development tools, code organization, and project presentation.

Technical Projects

AWS S3 to Redshift Data Lakehouse Project — *AWS S3, Redshift, PySpark, Hive, Scala, SQL, Parquet*

- Designed an end-to-end batch data engineering workflow using S3 raw, processed, and curated zones to organize structured and semi-structured datasets.
- Applied Python/PySpark-style and Scala Spark transformation logic for schema validation, null handling, deduplication, type casting, joins, aggregations, and curated dataset preparation.
- Modeled Redshift staging, fact, and dimension tables and wrote SQL validation checks for record counts, duplicates, schema rules, and source-to-target reconciliation.

Emotion-Aware Support Companion — *TypeScript, React, Node.js, Express.js, PostgreSQL, Prisma, REST APIs*

- Developed a full-stack web application with conversation sessions, message analysis, consent management, privacy preferences, feedback, and audit logging.
- Designed PostgreSQL/Prisma entities for users, sessions, messages, emotion records, crisis flags, privacy settings, feedback entries, and audit records.
- Validated functionality using Gherkin acceptance scenarios with Cucumber.js and Playwright, including UI, database, safety, and performance validations.

Personalized Content Recommendation Prototype — *Python, SQL, Pandas, NumPy, Matplotlib, ML Concepts*

- Built a recommendation-style analytics prototype using user-item interaction data to prepare features, analyze patterns, and support personalized content suggestions.
- Used Python and SQL concepts to clean datasets, validate records, calculate engagement metrics, and produce analysis-ready outputs for recommendation logic.

Brain Tumor Detection Using MRI Scan Images — *Python, TensorFlow, Keras, OpenCV, CNN, NumPy, Pandas*

- Built a CNN-based image classification workflow to classify MRI images as tumor or no tumor using preprocessing, train/validation/ splitting, and model evaluation.
- Performed image resizing, rescaling, augmentation, accuracy/loss tracking, results visualization, and analytical troubleshooting, achieving approximately 94% test accuracy in the project report.

Learning Management System Web Application — *Java, SQL Concepts, Web Design*

- Designed an academic web application concept to manage student information, course materials, assignments, and grades using OOP, database design, and SQL concepts.

Certifications

Fundamentals of Data Analytics | Microsoft Azure Fundamentals | Oracle Certified Foundations Associate | Aviatrix Certified Engineer – Multicloud Network Associate | Advanced RPA Professional | Introduction to Packet Tracer | OS Bridge Course